

Systemic Gender Inequities in Who Reviews Code

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April 27, 2026

Code review is a fundamental practice in modern software development, serving not only as a mechanism for ensuring code quality but also as a critical avenue for knowledge sharing and collaboration among engineers. In typical workflows, developers submit code changes and assign peers to review them, facilitating feedback, learning, and dissemination of project-specific knowledge across teams. Prior research has highlighted the importance of code review in improving code quality, enabling knowledge transfer, and fostering collective ownership of software systems. Given its central role, equitable participation in code review is essential to ensure that all developers benefit from and contribute to this collaborative process.

However, despite its importance, there is growing concern that participation in code review may not be evenly distributed across developers. This paper investigates systemic gender inequities in code review responsibilities within a large software organization, highlighting how such imbalances can impact both individuals and organizations. Unequal distribution of review work can limit opportunities for learning and visibility, while also reinforcing broader workplace disparities. Drawing on insights from management and workplace studies, the authors emphasize that inequitable task allocation can reduce productivity, hinder career advancement, and negatively affect organizational performance. Motivated by these concerns, the study aims to understand the extent, causes, and implications of gender-based disparities in code review participation.

1 Motivation

The motivation for this paper stems from the recognition that code review is not merely a quality assurance activity, but a **central mechanism for knowledge sharing, skill development, and collaboration** in modern software engineering. As highlighted in the introduction, code review enables engineers to learn new approaches, understand system architecture, and become familiar with shared codebases, thereby playing a critical role in both individual growth and team effectiveness. Given these benefits, equitable participation in code review is essential to ensure that all engineers have access to opportunities for learning and visibility. However, despite its importance, there has been limited investigation into whether these opportunities are distributed fairly across different demographic groups, particularly with respect to gender.

The paper is further motivated by broader findings from workplace and management literature, which consistently show that **work is not distributed evenly across employees**, and that demographic factors such as gender can significantly influence task allocation. Prior studies have demonstrated that men are more likely to be assigned **high-visibility, career-advancing tasks**, while women are more frequently tasked with **support-oriented or non-promotable work**. Such disparities are critical because promotable tasks contribute directly to career progression, performance evaluations, and skill development, whereas non-promotable tasks, although necessary, offer limited long-term benefits. This imbalance raises concerns that similar inequities may exist in software engineering practices such as code review.

Another key motivation lies in understanding the **consequences of inequitable knowledge-sharing opportunities**. The paper draws on prior research suggesting that unequal participation in collaborative tasks can lead to **reduced productivity, lower work quality, and diminished perceptions of competence at the individual level**, while at the organizational level, it can result in **decreased**

innovation and overall performance. Since code review is a primary channel through which knowledge is exchanged, any imbalance in participation can exacerbate existing inequalities, limiting both individual growth and organizational effectiveness.

Additionally, the authors are motivated by the observation that **systemic and structural factors may reinforce these inequities.** For instance, reviewer selection is often influenced by human decisions, such as authors choosing reviewers based on familiarity or perceived expertise, which may introduce bias. Furthermore, automated tools like reviewer recommender systems can unintentionally **amplify existing biases** if they are trained on historical data that reflects unequal participation. These dynamics suggest that inequities in code review are not merely incidental but may be embedded within the socio-technical systems that govern software development workflows.

Finally, the paper is driven by a practical goal: to move beyond identifying inequities and toward **designing interventions that promote fairness in collaborative software practices.** By quantifying disparities and analyzing their root causes, the study seeks to provide actionable insights that can help organizations create more equitable development environments. This includes rethinking how reviewers are selected, how expertise is recognized, and how workload is distributed, ultimately aiming to ensure that **code review fulfills its role as an inclusive and effective mechanism for knowledge sharing.**

2 Methodology

2.1 Data Source and Scope of Analysis

The study adopts a large-scale empirical approach by leveraging **extensive internal code review data from Google**, focusing specifically on reviews conducted through its primary review tool, Critique. To ensure consistency and relevance, the authors restrict their analysis to **submitted changelists (i.e., merged code changes)**, excluding incomplete or abandoned reviews that may not reflect actual participation patterns. This design choice helps capture meaningful contributions to code review activity rather than incidental or partial interactions.

Additionally, the dataset incorporates **pre-existing gender information from Google’s diversity reports**, which is noted to be more than 99% complete across employees. While the gender classification is binary (male and female), the authors explicitly acknowledge its limitations, ensuring transparency in how demographic data is used in the analysis. By combining behavioral data (code reviews) with demographic attributes, the study is positioned to quantitatively assess disparities in participation. The analysis is further contextualized by considering the organizational structure and culture of Google, which may influence how code review responsibilities are assigned and perceived. This comprehensive data source allows the researchers to conduct a robust examination of systemic inequities in code review participation.

2.2 Definition and Measurement of Review Load

A key methodological contribution lies in how the paper defines and operationalizes the concept of review workload. Instead of using queue length (i.e., pending reviews), the authors define review load as the **number of reviews performed by an engineer within a fixed time window.** This distinction is important because it captures actual participation rather than potential workload or availability.

The data is aggregated over specific periods (e.g., three months), allowing the researchers to construct a **cross-sectional view of review activity.** This aggregation helps smooth out short-term fluctuations and provides a stable basis for comparing participation across individuals and groups. By focusing on completed reviews, the study directly measures contribution to collaborative knowledge-sharing processes.

2.3 Quantitative Analytical Framework

The core of the methodology is a **quantitative, regression-based analysis** designed to examine the relationship between gender and review participation. In this framework, the primary dependent

variable is the number of code reviews performed, while the main independent variable of interest is gender. Regression models are employed to **control for confounding factors**, ensuring that observed disparities are not simply due to differences in experience, role, or other measurable attributes.

This statistical approach enables the authors to isolate the effect of gender on review load and to quantify disparities more rigorously. By using cross-sectional data and controlling for multiple variables, the study strengthens its ability to support **plausible causal interpretations** rather than simple correlations.

2.4 Identification of Antecedents

Beyond measuring disparities, the methodology is designed to uncover their underlying causes. The authors take a **holistic and exploratory approach to identifying potential antecedents** of inequity, considering multiple factors that could influence review participation. These include aspects such as how reviewer credentials (e.g., code ownership) are assigned, how authors select reviewers, and how automated systems recommend reviewers.

Each potential factor is analyzed using data-driven techniques, with the goal of determining whether it contributes to observed inequities. The researchers prioritize analyses that are **feasible with available data, enable causal reasoning, and yield actionable insights**. This systematic exploration allows the study to move beyond surface-level observations and provide a deeper understanding of systemic influences.

2.5 Consideration of Socio-Technical Factors

An important methodological aspect is the inclusion of both **human decision-making processes and automated system behaviors** in the analysis. For example, the study examines how manual reviewer selection by authors and algorithmic recommendations from tools interact to shape review participation patterns. This reflects a socio-technical perspective, recognizing that inequities may emerge from the **interaction between human biases and system design**.

By analyzing these dimensions together, the methodology captures the complexity of real-world software development environments, where both social and technical factors jointly influence outcomes. This approach strengthens the study's ability to explain not just whether inequities exist, but **how they are produced and sustained**.

2.6 Privacy and Ethical Considerations

Given the sensitive nature of employee data, the study adheres to strict privacy protocols. The researchers follow **established privacy principles for analyzing internal data** and ensure that the study focuses only on work-related activities. The research design is reviewed by an internal privacy working group, analogous to an Institutional Review Board, to ensure ethical compliance.

This careful handling of data underscores the study's commitment to responsible research practices, balancing the need for detailed analysis with the protection of individual privacy. It also enhances the credibility of the findings by demonstrating that the analysis is conducted within a well-governed ethical framework.

2.7 Evaluation of Interventions

Finally, the methodology extends beyond analysis to include the **evaluation of interventions aimed at reducing inequities**. The authors assess the impact of specific changes (e.g., adjustments in reviewer assignment processes) to determine whether they help close the gap in review participation. This experimental or quasi-experimental component adds a practical dimension to the study.

Rather than stopping at diagnosis, the paper emphasizes actionable outcomes by testing solutions in a real-world setting. This strengthens the contribution of the work, as it not only identifies systemic issues but also provides evidence on **how organizations can mitigate them effectively**.

3 Key Findings

3.1 Overall Inequity in Code Review Participation

One of the central findings of the paper is the existence of a clear and measurable gender disparity in code review participation. The authors find that **women perform approximately 25% fewer code reviews than men on average**, a gap that remains substantial even after accounting for confounding factors, reducing to about 16.8%. This demonstrates that the inequity is not merely a byproduct of differences in role, experience, or availability, but reflects a systemic imbalance in how review work is distributed.

This finding is particularly significant because code review is a key avenue for knowledge sharing and visibility. A lower participation rate implies that women may have **fewer opportunities to engage with diverse codebases, learn from peers, and build technical reputation**, which can have downstream effects on career growth and recognition within teams.

3.2 Human Selection Bias in Reviewer Assignment

A major contributing factor identified in the study is the role of human decision-making in reviewer selection. The authors find that **code authors tend to select men more frequently as reviewers**, even when other qualified reviewers are available. This preference introduces bias at the very first stage of the review assignment process.

This behavior reflects implicit biases or habitual selection patterns, where authors may rely on familiarity, perceived expertise, or prior interactions when choosing reviewers. As a result, certain groups - particularly women—may be systematically overlooked, reinforcing existing participation gaps. Over time, this leads to a **self-reinforcing cycle**, where those who are selected more often gain more visibility and experience, making them even more likely to be selected in the future.

3.3 Inequities in Reviewer Credentials and Expertise Recognition

Another key finding relates to how reviewer credentials, such as code ownership and expertise, are assigned and recognized. The study shows that there are **gender differences in how such credentials are earned and attributed**, which in turn affects reviewer selection.

Because reviewer recommendation and selection often depend on perceived expertise, any imbalance in credential assignment can directly translate into unequal opportunities. If women are less likely to be recognized as code owners or domain experts, they are also less likely to be invited to review code. This creates a structural barrier where **limited recognition leads to fewer opportunities, and fewer opportunities further limit recognition**, perpetuating inequity over time.

3.4 Amplification of Bias by Recommender Systems

The study also uncovers the role of automated systems in reinforcing disparities. Reviewer recommendation tools, which are designed to assist authors in selecting reviewers, can **amplify existing human biases when trained on historical data**.

Since these systems learn from past behavior, any pre-existing imbalance in reviewer selection is likely to be encoded into the model. As a result, the system may disproportionately recommend reviewers who have historically been chosen more often—typically men—thereby **scaling and institutionalizing bias through automation**. This finding highlights the risk of relying on data-driven systems without accounting for fairness and equity considerations.

3.5 Systemic Nature of the Inequity

A crucial insight from the findings is that the observed disparity is not due to a single factor but arises from the **interaction of multiple socio-technical mechanisms**. Human biases, structural differences in credentialing, and algorithmic reinforcement all contribute to the unequal distribution of review work.

This layered explanation underscores that the issue is systemic rather than incidental. The inequity is embedded in workflows, tools, and organizational practices, making it more difficult to address through isolated changes. The findings emphasize that **effective solutions must consider the broader ecosystem of code review practices rather than focusing on individual behaviors alone.**

3.6 Impact of Interventions

Finally, the paper provides evidence that targeted interventions can help mitigate these disparities. The authors report that even a relatively small change in the system or process can **begin to reduce the gap in review participation.**

While the paper does not claim to fully eliminate inequity, this result is important because it demonstrates that the system is **amenable to improvement through thoughtful design changes.** It suggests that organizations can take concrete steps—such as modifying reviewer recommendation strategies or encouraging more balanced selection practices—to move toward more equitable participation.

3.7 Broader Implications of the Findings

Taken together, the findings reveal that inequities in code review are not only measurable but also consequential. They affect who gains experience, who builds expertise, and who becomes visible within the organization. By highlighting these dynamics, the paper shows that code review is not just a technical process but a **social mechanism that shapes career trajectories and knowledge distribution.**

This reframing is critical, as it positions fairness in code review not as a peripheral concern but as a central issue for both individual development and organizational effectiveness.

4 Implications

Knowledge Sharing and Collaboration The findings highlight that inequities in code review participation directly impact how knowledge is distributed within an organization. Since code review is a primary mechanism for learning system architecture, APIs, and coding practices, unequal participation leads to **uneven access to knowledge-sharing opportunities.** This can limit the ability of under-represented groups to gain exposure to diverse parts of the codebase, ultimately affecting both individual skill development and team-wide collaboration. The paper emphasizes that achieving equity in review participation is essential to ensure that **knowledge transfer benefits all engineers rather than a select subset.**

Individual Career Growth The study suggests that disparities in review participation can have significant consequences for career advancement. Code review contributes to building technical expertise, visibility, and reputation within teams. When certain groups participate less, they may have **fewer opportunities to demonstrate competence and gain recognition,** which can negatively influence performance evaluations and promotion prospects. Drawing from management literature, the paper reinforces that inequitable task distribution can **signal lower competence and reduce perceived contributions,** even when actual ability is comparable.

Organizational Performance At an organizational level, the inequitable distribution of review work can reduce overall effectiveness. The paper notes that unequal knowledge sharing and participation can lead to **lower productivity, reduced work quality, and diminished innovation.** When knowledge is concentrated among a subset of engineers, teams may become less resilient and less capable of handling complex or distributed tasks. Ensuring equitable participation in code review is therefore not only a fairness concern but also a **strategic requirement for maintaining high-performing and innovative engineering teams.**

Tool and System Design A key implication of the study is the need to rethink how socio-technical systems, such as reviewer recommendation tools, are designed. Since these systems can **amplify existing human biases,** organizations must incorporate fairness considerations into their design and evaluation.

This includes developing mechanisms that explicitly account for workload balance and equitable participation, rather than relying solely on historical data. The findings suggest that **algorithmic systems should be designed to mitigate, rather than reinforce, existing inequities.**

Organizational Practices and Policies The paper also points to the importance of organizational practices in shaping equitable participation. Reviewer selection processes, recognition of expertise, and assignment of responsibilities all influence who participates in code review. Organizations may need to adopt policies that encourage **more balanced reviewer selection and equitable distribution of tasks**, ensuring that opportunities are not concentrated among a limited group of engineers. This could involve guidelines, training, or process changes that make equity an explicit consideration in everyday workflows.

Intervention and System Improvement Finally, the study demonstrates that targeted interventions can begin to address inequities, suggesting that the problem is **actionable rather than immutable**. Even small changes to systems or processes can reduce disparities, indicating that organizations have the ability to improve equity through deliberate design choices. This reinforces the importance of continuous monitoring and iterative improvement, where organizations evaluate the impact of interventions and refine them over time to achieve more equitable outcomes.